Chapter Summary

We discussed various types of first-order DE in this chapter: Separable, linear, and exact DE. Their principal features and method of solution are outlined below.

- **Separable Equations:** $dy/dx = g(x)p(y)$.
  Separate the variables and integrate.

- **Linear Equations:** $dy/dx + P(x)y = Q(x)$,
  The integrating factor $\mu = e^{\int p(x)\ dx}$ reduces the equation to $d[\mu y]/dx = \mu Q$ so that $\mu y = \int \mu Q dx + C$.

- **Exact Equations:** $dF(x.y) = 0$. Solutions are given implicitly by $F(x,y) = c$. If $\partial M/\partial y = \partial N/dx$, then $Mdx + Ndy = 0$ is exact and F is given by

  $$F = \int Mdx + g(y), \text{ where } g'(y) = N - \frac{\partial}{\partial y}\int Mdx$$

  or

  $$F = \int Ndy + h(x), \text{ where } h'(x) = M - \frac{\partial}{\partial x}\int Ndy$$

  When an equation is not separable, linear, or exact, it may be possible to find an integrating factor or perform a substitution that will enable us to solve the equation with methods we've learned.

- **Integration factors:** $\mu M\ dx + \mu N\ dy = 0$ **is exact.** If $(\partial M/\partial y - \partial N/\partial x)/N$ depends only on x, then

$$\mu(x) = e^{\int (\frac{\partial M/\partial y - \partial N/\partial x}{N})\ dx}$$

is an integrating factor. If $(\partial N/\partial x - \partial M/\partial y)/M$ depends only on y, then

$$\mu(y) = e^{\int (\frac{\partial N/\partial x - \partial M/\partial y}{M})\ dy}$$

is an integrating factor.

- **Homogeneous Equations:** $dy/dx = G(y/x)$. Let $v = y/x$. Then $dy/dx = v + x(dv/dx)$, and the transformed equation in the variables v and x is **separable**.

- **Bernoulli Equations:** $dy/dx + P(x)y = Q(x)y^n$. For $n \neq 0$ or 1, let $v = y^{1-n}$. Then $dv/dx = (1-n)y^{-n}(dy/dx)$, and the transformed equation in the variables v and x is **linear**. $\frac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x)$

- **Equations of the Form:** $dy/dx = G(ax + by)$. Let $z = ax + by$. Then $dz/dx = a + b(dy/dx)$, and the transformed equation in the variables z and x is **separable**.

Existence and Uniqueness

Given an IVP $\frac{dy}{dx} = f(x,y)$, $y(x_0) = y_0$ (x_0, y_0)

If $f(x,y)$ and $f_y(x,y)$ are continuous "around" (x_0, y_0) then the DE has a unique solution through (x_0, y_0).

dep

Ex: Consider $\frac{dy}{dx} = 3y^{2/3}$, $y(0) = 1$. Does this have a unique solution around $x = 0$?

x y
(0, 1)

$f(x,y) = 3y^{2/3}$

$f_y(x,y) = 2y^{-1/3}$

$f(0,1) = 3(1)^{2/3} = 3$ exists ✓

$f_y(0,1) = 2(1)^{-1/3} = 2$ exists

So there is a unique solution. Only when $y = 0$ are we <u>not</u> guaranteed a unique solution.

Euler's Method

Given an IVP $\frac{dy}{dx} = f(x,y)$, $y(x_0) = y_0$

Given a step size h, we can approximate outputs with

$$\begin{cases} x_{n+1} = x_n + h \\ y_{n+1} = y_n + h f(x_n, y_n) \end{cases}$$

Ex: Consider $\frac{dy}{dx} = x - y^2$, $y(1) = 2$ $\quad x_0 = 1$, $y_0 = 2$

Approximate $y(2)$ using Euler's Method with step size $h = 0.5$.

$f(x,y)$

step 1
$$\begin{cases} x_1 = x_0 + h = 1 + 0.5 = 1.5 \\ y_1 = y_0 + hf(x_0, y_0) = 2 + 0.5(1 - (2)^2) \end{cases}$$
$$= 2 + 0.5(-3)$$
$$= 2 - 1.5 = 0.5$$

$(x_1, y_1) = (1.5, 0.5)$

step 2
$$\begin{cases} x_2 = x_1 + h = 1.5 + 0.5 = 2 \\ y_2 = y_1 + h\, f(x_1, y_1) = 0.5 + 0.5(1.5 - (0.5)^2) \end{cases}$$
$$= 0.5 + 0.5(1.5 - 0.25)$$
$$= 0.5 + 0.5(1.25)$$
$$= 0.5 + 0.625$$
$$= \boxed{1.125}$$

Ex: Solve $\frac{dy}{dx} + y\tan(x) = -\sin(x)$ linear!

$p(x)$ $\quad Q$

use integrating factor $\mu(x) = e^{\int P(x)dx}$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$\Rightarrow \mu(x) = e^{\int \tan(x)dx} = e^{\ln|\sec(x)|} = \sec(x)$$

Recall, $y = \frac{1}{\mu(x)}\left[\int \mu(x)\,Q(x)\,dx + C\right]$

$$= \frac{1}{\sec(x)}\left[\int \sec(x)(-\sin(x))\,dx + C\right]$$

$$= \cos(x)\left[\int -\tan(x)\,dx + C\right]$$

number

e^C

$$= \cos(x)\left[-\ln|\sec(x)| + C\right]$$

$$= \cos(x)\left[\ln|\cos(x)| + C\right]$$

$$= \cos(x)\ln|\cos(x)| + C\cos(x)$$

function

Ex: Solve $\frac{dx}{dt} = 1 + \cos^2(\underbrace{t-x}_{\text{linear } ax+bt})$ — $G(ax+by+c)$ — write in implizit form

use $z = t - x \Rightarrow \frac{dz}{dt} = 1 - \frac{dx}{dt} \Rightarrow \frac{dx}{dt} = 1 - \frac{dz}{dt}$

$\Rightarrow \cancel{1} - \frac{dz}{dt} = \cancel{1} + \cos^2(z)$ separable

$\Rightarrow -\frac{dz}{dt} = \cos^2(z)$

$\Rightarrow \int \frac{1}{\cos^2(z)}\,dz = \int -dt \Rightarrow \int \sec^2(z)\,dz = -t + C$

$\Rightarrow \tan(z) = -t + C$

$\Rightarrow \tan(t-x) = -t + C$

$\Rightarrow \boxed{\tan(t-x) + t = C}$

Ex: Solve $\frac{dy}{dx} \underbrace{- 2x^{-1}}_{P(x)} y = \underbrace{x^{-1}}_{Q(x)} y^{-1}$ $\longrightarrow$ Bernoulli explicit form

power of y suppose

here $y^{-1} \Rightarrow n=-1$, $1-n = 2$, $v = y^2$ (y^{1-n}) $\underline{\underline{y(1)=3}}$

$$\Rightarrow \frac{dv}{dx} + \underbrace{(2)}_{1-n}\underbrace{(-2x^{-1})}_{P}\underbrace{v}_{v} = \underbrace{2 \cdot x^{-1}}_{1-n\ Q}$$

$$\Rightarrow \frac{dv}{dx} - \frac{4}{x} v = \frac{2}{x}$$ linear!

$$\mu(x) = e^{\int -\frac{4}{x} dx} = e^{-4 \ln(x)} = x^{-4}$$

$$\Rightarrow v = \frac{1}{x^{-4}} \left[\int x^{-4}(2x^{-1}) dx + C \right]$$

$$\Rightarrow v = x^4 \left[\int 2x^{-5} dx + C \right] = x^4 \left[-\frac{1}{2} x^{-4} + C \right]$$

$$\Rightarrow v = -\frac{1}{2} + Cx^4$$

$$\Rightarrow y^2 = Cx^4 - \frac{1}{2} \Rightarrow y = \pm\sqrt{Cx^4 - \frac{1}{2}}$$

If $y(1) = 3$ $\Rightarrow y = \sqrt{Cx^4 - \frac{1}{2}}$

positive

$$3 = \sqrt{C - \frac{1}{2}} \Rightarrow 9 = C - \frac{1}{2} \Rightarrow C = \frac{19}{2}$$

Thus $\boxed{y = \sqrt{\frac{19}{2}x^4 - \frac{1}{2}}}$

Ex: Consider $\underbrace{y(x-y-2)}_{M}dx + \underbrace{x(y-x+4)}_{N}dy = 0$

$M_y = 1(x-y-2) + y(-1) = x-2y-2$ $\neq$ not exact.

$N_x = 1(y-x+4) + x(-1) = y-2x+4$

$$\frac{M_y - N_x}{N} = \frac{(x-2y-2)-(y-2x+4)}{x(y-x+4)} = \frac{3x-3y-6}{x(y-x+4)}$$

terms of x

$$= \frac{3(x-y-2)}{x(y-x+4)} \times$$

try $$\frac{N_x - M_y}{M} = \frac{(y-2x+4)-(x-2y-2)}{y(x-y-2)} = \frac{-3x+3y+6}{y(x-y-2)}$$

terms of y

$$= \frac{-3\cancel{(x-y-2)}}{y\cancel{(x-y-2)}} = \frac{-3}{y} \checkmark$$

so $\mu(y) = e^{\int \frac{-3}{y}dy} = e^{-3\ln|y|} = y^{-3}$

multiply by y^{-3}: $y^{-3}[y(x-y-2)dx + x(y-x+4)dy = 0]$

$\Rightarrow y^{-2}(x-y-2)dx + xy^{-3}(y-x+4)dy = 0$

$\Rightarrow (xy^{-2} - y^{-1} - 2y^{-2})dx + (xy^{-2} - x^2y^{-3} + 4xy^{-3})dy = 0$

is exact. $f_x = M$, $f_y = N$

new

$$f = \int f_x\,dx = \int M\,dx = \int (xy^{-2} - y^{-1} - 2y^{-2})\,dx + g(y)$$

$$= \tfrac{1}{2}x^2y^{-2} - xy^{-1} - 2xy^{-2} + g(y)$$

$$xy^{-2} - x^2y^{-3} + 4xy^{-3} = N = f_y = \frac{\partial}{\partial y}\left(\tfrac{1}{2}x^2y^{-2} - xy^{-1} - 2xy^{-2} + g(y)\right)$$

$$= -x^2y^{-3} + xy^{-2} + 4xy^{-3} + g'(y)$$

$$\Rightarrow g'(y) = 0 \Rightarrow g(y) = 0.$$

So $f(x,y) = \boxed{\tfrac{1}{2}x^2y^{-2} - xy^{-1} - 2xy^{-2} = C}$

<u>Curvature</u> if $y = f(x)$

$$K = \frac{|y''|}{[1+(y')^2]^{3/2}} \quad \left(K = \frac{|\vec{r}' \times r''|}{|r'|^3}\right)$$

function

curvature of $x^2+y^2=r^2$ is $K = \frac{1}{r}$

Suppose $\frac{1}{r} = \frac{|y''|}{[1+(y')^2]^{3/2}}$

second order y'', y'

$\Rightarrow r\,y'' = [1+(y')^2]^{3/2}$

set $v = y'$

$\Rightarrow r\,v' = [1+v^2]^{3/2}$

first order

$\Rightarrow r\frac{dv}{dx} = [1+v^2]^{3/2}$

no x's separable

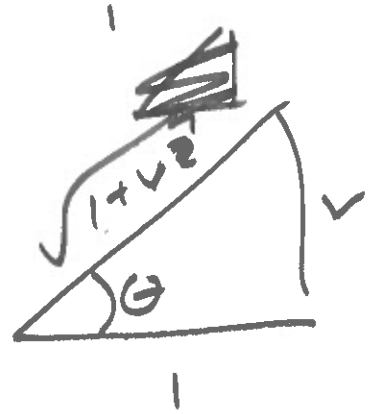
$v = \tan\theta$

$dv = \sec^2\theta\,d\theta$

$$\int \frac{r}{[1+v^2]^{3/2}}\,dv = \int dx$$

$$\int \frac{r\sec^2\theta}{[\sec^2\theta]^{3/2}}\,d\theta = x + C$$

$\frac{v}{1} = \tan\theta$



$$\Rightarrow \int r\cdot\frac{1}{\sec\theta}\,d\theta = x + C$$

$$\Rightarrow \int r\cos\theta\,d\theta = r\sin\theta = x + C$$

$$\Rightarrow r\cdot\frac{v}{\sqrt{1+v^2}} = x + C$$

$$\Rightarrow r\cdot\frac{y'}{\sqrt{1+(y')^2}} = x + C$$

$$\Rightarrow r\cdot y' = (x+C)\sqrt{1+(y')^2}$$

$$\Rightarrow r^2(y')^2 = (x+C)^2(1+(y')^2) = (x+C)^2 + (x+C)^2(y')^2$$

$$\Rightarrow [r^2-(x+C)^2](y')^2 = (x+C)^2$$

$$\Rightarrow r(y') = \sqrt{\frac{(x+C)^2}{r^2-(x+C)^2}} = \frac{x+C}{\sqrt{r^2-(x+C)^2}}$$

$x + C = r\sin\theta$

$dx = r\cos\theta\,d\theta$

$$\Rightarrow y = \frac{1}{r}\int \frac{x+C}{\sqrt{r^2-(x+C)^2}}\,dx$$

$$= \sqrt{r^2-(x+C)^2} + D$$

$$\Rightarrow y - D = \sqrt{r^2-(x+C)^2}$$

$$\Rightarrow (y-D)^2 = r^2-(x+C)^2$$

$$\Rightarrow (x+C)^2 + (y-D)^2 = r^2$$ <u>circle</u>